$$\boxed{v\cdot(Au)=(v\cdot A)u}$$

1. $(Au)\cdot v=(A\cdot v)u$, $A\in R^{m\times n}$, $u\in R^n$, $v\in R^m$

(here $\cdot$ is $\dot{e}$)

proof: $(Au)\cdot v=\begin{pmatrix}a_1^Tu\\ a_m^Tu\end{pmatrix}\cdot v$

$$=\begin{pmatrix}(v_1a_1)^Tu\\ (v_ma_m)^Tu\end{pmatrix}$$

$$=\begin{pmatrix}(v_1a_1)^T\\ (v_ma_m)^T\end{pmatrix}u$$

$$=(A\cdot v)u$$

$\therefore (Au)\cdot v=v\cdot(Au)=(A\cdot v)u=(v\cdot A)u$

2. $(Au)\cdot v\neq A(u\cdot v)$, if $m=n$.

proof: (note $u\cdot v$ is meaningless if $m\neq n$)

$\because (Au)\cdot v=\begin{pmatrix}a_1^Tuv_1\\ \cdots\end{pmatrix}$, $A(u\cdot v)=\begin{pmatrix}a_1^T(u\cdot v)\\ \cdots\end{pmatrix}$

$a_1^Tuv_1\neq a_1^T(u\cdot v)$

$\therefore (Au)\cdot v\neq A(u\cdot v)$